

Perceived Monetary Policy Rule: Identification and Macroeconomic Implications

Husang Kim ¹

¹Department of Economics
University of Texas at Austin

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Motivation and Research Question

1. Taylor rule shows how actively monetary policy responds:

$$i_t = \underbrace{\phi_{\pi,t}}_{\text{Rule}} \pi_t$$

2. FIRE: Perfect knowledge of the Fed's rule.

$$\phi_{\pi,t|t} = \phi_{\pi,t}$$

3. Distinct role of **perceived rule** ($\phi_{\pi,t|t} \neq \phi_{\pi,t}$) underexplored.

Research Question

1. *What are the macroeconomic implications of the perceived rule ($\neq$ actual rule)?*
2. *How do we identify the perceived rule ($\phi_{\pi,t|t}$)?*

Overview - Identification

1. Identifying the bias in **perceived**-rule estimation

- *Convention*: Run Taylor-rule regression using interest rate forecasts.
- *Bias*: Forecasts embed uncertainty about the future rule, biasing ϕ_π downward.
- *Identification*: **✗** Interest rate forecast-based **✓** Nowcast-based.
- *Nowcast*: Real-time projection of the *current* period value.

2. Restoring the **perceived** Taylor principle (1983-2024)

- *Taylor principle*: Raise i more than 1:1 with π .
- *Forecast-based*: Taylor principle **✗** ($\phi_{\pi,t|t} < 1$).
- *Nowcast-based*: Taylor principle **✓** ($\phi_{\pi,t|t} > 1$).

Overview - Macroeconomic Implications

Subjective Perception Markov Switching New Keynesian (SP-MSNK) model.

Perception ($\phi_{\pi,t|t}$) vs. Direct Policy ($\phi_{\pi,t}$)

1. Reorienting the Taylor principle to **perception**

- Determinacy depends on whether agents **believe** in it, not the actual rule.
- ✗ Implement the Taylor principle ✓ Anchor **perception** to it.

2. Reinterpreting the source of Great Moderation (1980-90s)

- ✗ The shift in monetary policy regime ✓ The shift in **perception**.
- Without it, peak inflation 4.8% → 7.4% and volatility $\times 1.93$.

3. Quantifying the cost of misreading **perception** (2021-23)

- Optimal lift-off timing in 2021-2023?
- **N**-based (2021Q1) vs. **F**-based (2021Q3).

Contribution to the literature

1. **Macroeconomic effects of policy perceptions:** Bauer, Pflueger, and Sunderam 2024, Schorfheide 2005, Hamilton, Pruitt, and Borger 2011, Bocola et al. 2024. . .
⇒ Evaluate GE consequences for when perceived rules differ from actual rules
2. **Estimating perceived monetary policy rules:** Hamilton, Pruitt, and Borger 2011; Bundick 2015; Jia, Shen, and Zheng 2023; Bocola et al. 2024; Bauer, Pflueger, and Sunderam 2024. . .
⇒ Identify structural covariance bias and propose nowcast-based identification
3. **Determinacy and the Great Moderation:** Lubik and Schorfheide 2004, Coibion and Gorodnichenko 2011, Bianchi 2013, Sims and Zha 2006. . .
⇒ Reinterpret Great Moderation as driven by “Good Perceived Policy”
4. **Regime-switching DSGE models:** Davig and Leeper 2007; Bianchi 2013; Barthélemy and Marx 2017; Bianchi, Lettau, and Ludvigson 2022. . .
⇒ Relax FIRE assumption to isolate the perception channel.
5. **Post-pandemic policy and credibility:** Nakamura, Riblier, and Steinsson 2025, Bocola et al. 2025. . .
⇒ Show perceived Taylor principle expands the flexibility of the central bank.

Perceived Rule vs. Actual Rule - Simple Model

Actual Model

$$i_t = E_t^P[\pi_{t+1}] + r_t$$

$$r_t = \rho_r r_{t-1} + v_t$$

$$i_t = \phi_f \pi_t$$

Subjective Model

$$i_t = E_t^P[\pi_{t+1}] + r_t$$

$$r_t = \rho_r r_{t-1} + v_t$$

$$i_t = \phi_p \pi_t$$

- $E_t^P[\cdot]$: private sector expectation under its **subjective** model.
- Agent observes r_t perfectly, but $\phi_p \neq \phi_f$.
- FIRE is the special case $\phi_p = \phi_f$, where $E_t^P[\cdot] = \mathbb{E}_t[\cdot]$.

Perceived Rule vs. Actual Rule - Simple Model

Perceived Law of Motion

$$i_{t+h|t} = \rho_r^h \underbrace{\frac{\phi_p}{\phi_p - \rho_r} r_t}_{i_{t|t}}$$

$$\pi_{t+h|t} = \rho_r^h \underbrace{\frac{1}{\phi_p - \rho_r} r_t}_{\pi_{t|t}}$$

Actual Law of Motion

$$i_t = \frac{\phi_p}{\phi_p - \rho_r} r_t$$

$$\pi_t = \underbrace{\frac{\phi_p}{\phi_f} \left[\frac{1}{\phi_p - \rho_r} r_t \right]}_{\text{FIRE } (\phi_p = \phi_f)} = \frac{1}{\phi_f} \underbrace{\left(\frac{\rho_r}{\phi_p - \rho_r} r_t \right)}_{E_t^P[\pi_{t+1}]} + \frac{1}{\phi_f} r_t$$

Takeaways

1. Determinacy: *perceived* Taylor principle ($\phi_p > 1$), NOT $\phi_f > 1$.
2. Perception Channel (through $E_t^P[\pi_{t+1}]$) vs. Direct Policy channel
3. Regressing $i_{t+h|t}$ on $\pi_{t+h|t}$ recovers ϕ_p , at any h .

Identification of Perceived Rule

Conventional practice regresses interest rate *forecasts* on inflation *forecasts*:
Hamilton, Pruitt, and Borger (2011) and Bauer, Pflueger, and Sunderam (2024)

$$E_t[i_{t+h}] = \beta_{OLS,IV} E_t[\pi_{t+h}] + e_{th} \quad \Rightarrow \quad \hat{\beta}_{OLS,IV} = \phi_p$$

But the monetary policy rule could move over time:

Kim and Nelson (2006), Coibion and Gorodnichenko (2011), and Bianchi (2013)

$$i_t = \phi_{\pi,t} \pi_t \Rightarrow E_t[i_{t+h}] = E_t[\phi_{\pi,t+h}] E_t[\pi_{t+h}] + \overbrace{\text{Cov}_t(\pi_{t+h}, \phi_{\pi,t+h})}^{\text{OVB}}$$

Takeaways

1. The covariance implies *downward* bias.
2. This does not depend on $E_t[\cdot] \neq \mathbb{E}_t[\cdot]$ or Perceived $\neq$ Actual.

Source of Covariance Bias - Illustration

Lubik and Schorfheide (2004) and Farmer, Waggoner, and Zha (2011):

$$x_t = \mathbb{E}_t x_{t+1} - \tau (i_t - \mathbb{E}_t \pi_{t+1})$$

$$\pi_t = \beta \mathbb{E}_t \pi_{t+1} + \kappa x_t + z_{S,t},$$

$$i_t = \phi_{\pi, s_t} \pi_t.$$

where $z_{S,t}$ follow persistent AR(1) process.

$$\phi_{\pi, s_t} \in \{\phi_{\pi, 1}, \phi_{\pi, 2}\}, \phi_{\pi, 1} > \phi_{\pi, 2} \text{ with } P = \begin{bmatrix} p & 1-p \\ 1-q & q \end{bmatrix}$$

The solution of this model is

$$\pi_t = \begin{cases} a_{1,S} z_{S,t} & \text{if } s_t = 1 \\ a_{2,S} z_{S,t} & \text{if } s_t = 2 \end{cases}$$

Source and Sign of Covariance Bias

At time t , with $s_t = j \in \{1, 2\}$:

$$\text{COV} [\phi_{\pi, s_{t+h}}, \pi_{t+h} \mid s_t = j] = \underbrace{[P^h]_{j1} (1 - [P^h]_{j1}) (\phi_{\pi, 1} - \phi_{\pi, 2})}_{\text{Regime uncertainty (+)}} \times \underbrace{(a_{1,s} - a_{2,s})}_{\text{Rate feedback, GE (-)}} \rho_S^h z_{St}$$

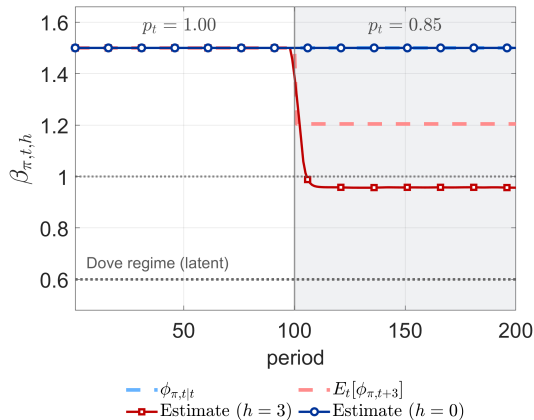
When is it nonzero?

- *Rate feedback*: zero if Partial Equilibrium.
- *Uncertainty*: zero if $\phi_{\pi, 1} = \phi_{\pi, 2}$ or $P = I$.

Sign of Bias

- $z_{St} > 0 \Rightarrow \mathbb{E}_t[\pi] > 0, \mathbb{C}_t < 0.$
- $z_{St} < 0 \Rightarrow \mathbb{E}_t[\pi] < 0, \mathbb{C}_t > 0.$

Numerical Illustration - Forecast vs. Nowcast



- Always Hawk: $\phi_{\pi,t|t} = 1.50$.
- But starts to doubt: $E_t[\phi_{\pi,t+3}] = 1.20$.

TVP estimation:

$$E_t[i_{t+h}] = \beta_{\pi,t,h} E_t[\pi_{t+h}] + e_{t,h}$$

- ✗ Forecast ($h = 3$): $0.96 < 1$.
- ✓ Nowcast ($h = 0$): 1.50.

Additional Remarks

1. Instrumental variables do not work

- The covariance is endogenous in every shock of the **subjective** model.

2. **Nowcast** identification is broader than the covariance

- The covariance says why **forecasts** fail, not what **nowcasts** require.
- Main assumption: the nowcast reflects the agent's **subjective** model.

3. $\mathbb{E}_t[\phi_{\pi,t+h}]$ vs. $\phi_{\pi,t|t}$: **distinct here, identical in the literature's practice**

- Constant rule, random walk, horizon-common coefficient: all equate them.

Empirical Framework

1. Forward-looking Taylor rule specification:

$$\begin{cases} i_t = (1 - \rho_t) \left[\phi_{\pi,t} \pi_{t+1|t}^{GB} + \phi_{x,t} (-2 \cdot u_{t+1|t}^{GB}) \right] + \rho_{1,t} i_{t-1} + \rho_{2,t} i_{t-2} + c_t + \varepsilon_t \\ i_{t|t} = (1 - \rho_{t|t}) \left[\phi_{\pi,t|t} \pi_{t+1|t}^{BC/SPF} + \phi_{x,t|t} (-2 \cdot u_{t+1|t}^{BC/SPF}) \right] + \rho_{1,t|t} i_{t-1|t} + \rho_{2,t|t} i_{t-2|t} + \tilde{c}_t + e_t \\ i_{t+h|t} = (1 - \rho_{t,h}) \left[\beta_{\pi,t,h} \pi_{t+h+1|t}^{BC/SPF} + \beta_{x,t,h} (-2 \cdot u_{t+h+1|t}^{BC/SPF}) \right] + \rho_{1,t,h} i_{t+h-1|t} + \rho_{2,t,h} i_{t+h-2|t} + \tilde{c}_{t,h} + e_{t,h} \end{cases}$$

Use Stock and Watson (1998) and Boivin (2005) + Stochastic Volatility.

2. Data:

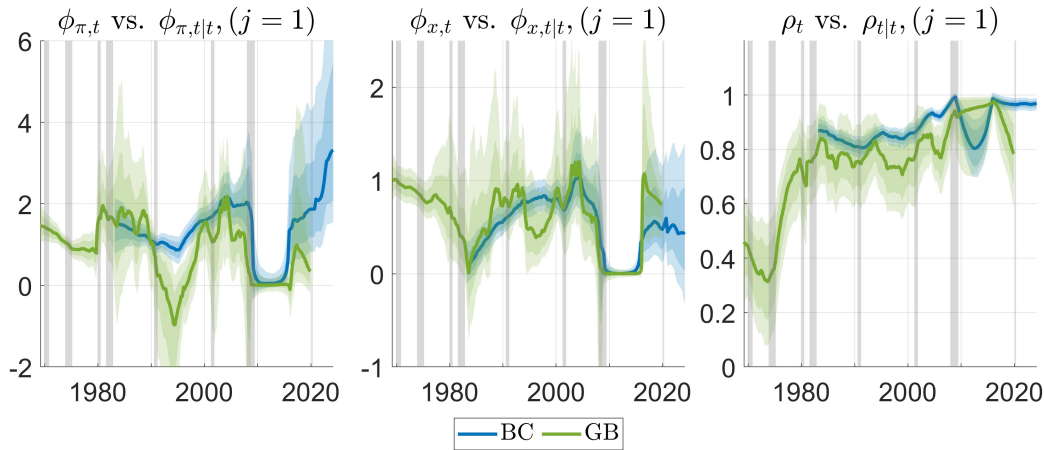
Actual Rule

- *Source:* Greenbook (Tealbook).
- *Sample:* 1969-2019, FOMC frequency.

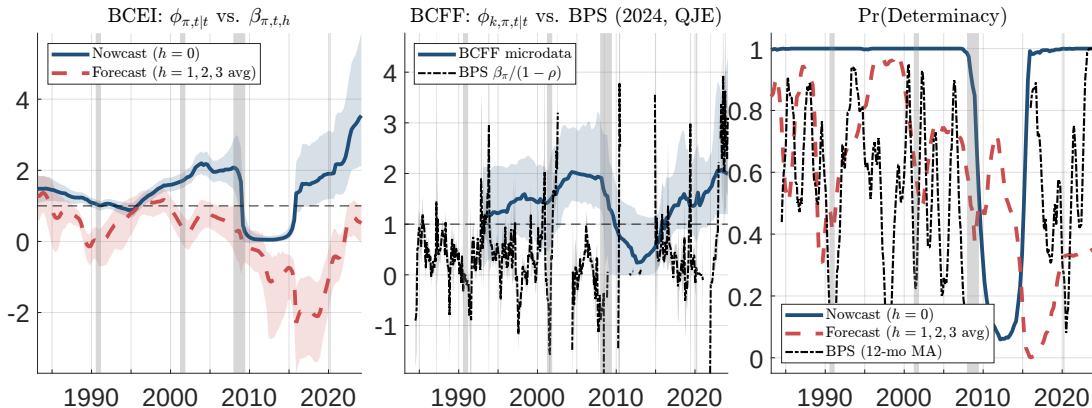
Perceived Rule

- *Baseline:* BCEI + SPF, 1983-2024, quarterly.
- *Robustness:* BCFF microdata.

Perceived vs. Actual Rule



Restoration of Perceived Taylor Principle



SP-MSNK Model: Framework and Solution

$$\left. \begin{array}{l} \text{Perceived } s_t^p \in \{1 \cdots S_p\} \text{ under subjective } \widehat{\mathbf{P}} \\ \text{Actual } s_t^f \in \{1 \cdots S_f\} \text{ under } \mathbf{P} \end{array} \right\} \implies \text{True state } \ell = (s_t^p, s_t^f)$$

1. **Perceived Law of Motion.** Agents believe, and solve, given $s_t^p = j$:

$$\Gamma_{0,j}^P E_t^P[\mathbf{y}_{t+1}] + \Gamma_{1,j}^P \mathbf{y}_t + \Gamma_{2,j}^P \mathbf{y}_{t-1} + \Gamma_{3,j}^P \boldsymbol{\varepsilon}_t^p = \mathbf{0} \implies \mathbf{y}_{t|t} = \widehat{\mathbf{A}}_j^{\text{PLM}} \mathbf{y}_{t-1} + \widehat{\mathbf{B}}_j^{\text{PLM}} \boldsymbol{\varepsilon}_t^p$$

2. **Expectations.** From that PLM, for $s_t^p = j$:

$$E_t^P[\mathbf{y}_{t+1}] = \sum_{k=1}^{S_p} \widehat{P}_{jk} \widehat{\mathbf{A}}_k^{\text{PLM}} \left(\widehat{\mathbf{A}}_j^{\text{PLM}} \mathbf{y}_{t-1} + \widehat{\mathbf{B}}_j^{\text{PLM}} \boldsymbol{\varepsilon}_t^p \right)$$

3. **Actual Law of Motion.** Substituting $E_t^P[\cdot]$, the economy obeys, given ℓ :

$$\Gamma_{0,\ell}^A E_t^P[\mathbf{y}_{t+1}] + \Gamma_{1,\ell}^A \mathbf{y}_t + \Gamma_{2,\ell}^A \mathbf{y}_{t-1} + \Gamma_{3,\ell}^A \boldsymbol{\varepsilon}_t = \mathbf{0} \implies \mathbf{y}_t = \mathbf{A}_\ell^{\text{ALM}} \mathbf{y}_{t-1} + \mathbf{B}_\ell^{\text{ALM}} \boldsymbol{\varepsilon}_t$$

Implication 1 – Determinacy depends on perception

Model ($S_p = S_f = 1$)

$$x_t = E_t^P [x_{t+1}] - \tau \left(i_t - E_t^P [\pi_{t+1}] - r_t^n \right)$$

$$\pi_t = \beta E_t^P [\pi_{t+1}] + \kappa x_t$$

Rules

Perceived: $i_t = r_t^n + \phi_\pi^P \pi_t$

Actual: $i_t = r_t^n + \phi_\pi^F \pi_t$

Under FIRE ($\phi_\pi^P = \phi_\pi^F$)

$$\pi_t = x_t = 0 \Rightarrow / \not\Leftarrow i_t = r_t^n$$

$$\Rightarrow / \Leftarrow i_t = r_t^n + \phi_\pi^F \pi_t$$

Peg (Galí 2015), then $\phi_\pi^F > 1$.

Dropping FIRE ($\phi_\pi^P \neq \phi_\pi^F$)

$$\pi_t = x_t = 0 \Rightarrow / \Leftarrow i_t = r_t^n$$

$$\text{Determinacy} \iff \phi_\pi^P > 1$$

Peg restored, without $\phi_\pi^F > 1$.

Determinacy is required to attain the target allocation $\pi_t = x_t = 0$.

Policy and empirical implications: Not $\phi_\pi^F > 1$, but $\phi_\pi^P > 1$.

SP-MSNK Model: Setup

IS and Phillips curve

$$x_t = E_t^P [x_{t+1}] - \tau (i_t - E_t^P [\pi_{t+1}]) + z_{Dt}$$

$$\pi_t = \beta E_t^P [\pi_{t+1}] + \kappa x_t + z_{St}$$

Rules ($S_p = S_f > 1$)

$$\text{Perceived: } i_t = \left(1 - \rho_{s_t^p}\right) \left[\phi_{\pi, s_t^p} E_t^P [\pi_{t+1}] + \phi_{x, s_t^p} E_t^P [x_{t+1}] \right] + \rho_{1, s_t^p} i_{t-1} + \rho_{2, s_t^p} i_{t-2}$$

$$\text{Actual: } i_t = \left(1 - \rho_{s_t^f}^F\right) \left[\phi_{\pi, s_t^f}^F E_t^P [\pi_{t+1}] + \phi_{x, s_t^f}^F E_t^P [x_{t+1}] \right] + \rho_{1, s_t^f}^F i_{t-1} + \rho_{2, s_t^f}^F i_{t-2} + \varepsilon_{Mt}$$

- s_t^p evolves under $\widehat{\mathbf{P}}$, s_t^f under $\mathbf{P}$, with $\mathcal{S}^p = \mathcal{S}^f = \mathcal{S} = \{1, \dots, S\}$.
- The Fed responds to E_t^P : to isolate the role of perceived vs. actual rules.

Counterfactual: Method

Given a sequence of $\mathbf{A}_\ell^{\text{ALM}}, \mathbf{B}_\ell^{\text{ALM}}$ for each t :

1. Recover structural shocks $\boldsymbol{\varepsilon}_t = [\varepsilon_{D,t}, \varepsilon_{S,t}, \varepsilon_{M,t}]'$ from data using baseline ALM:

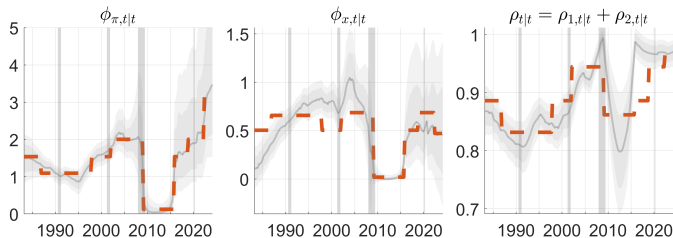
$$\mathbf{y}_t = \mathbf{A}_\ell^{\text{ALM}} \mathbf{y}_{t-1} + \mathbf{B}_\ell^{\text{ALM}} \boldsymbol{\varepsilon}_t$$

2. Construct and replace with counterfactual ALM: $(\mathbf{A}_{\ell^{\text{CF}}}^{\text{ALM}}, \mathbf{B}_{\ell^{\text{CF}}}^{\text{ALM}})$

t	$\dots$	T_1	$\dots$	T_2	$\dots$
$\ell = (s_t^p, s_t^f)$	$\dots$			(H, H)	$\dots$
ℓ^{CF1}	$\dots$			$(\mathbf{D}, H)$	$\dots$
ℓ^{CF2}	$\dots$		(H, H) but with different $\hat{P}$		$\dots$

3. Feed *same shocks* through counterfactual ALM $\Rightarrow$ counterfactual paths.

Calibration



- Univariate HMM on $\phi_{\pi,t|t}$, $K = 5$ by BIC $\Rightarrow (s_t^p)_{t=1}^T$.
- Other coefficients: within-regime averages.
- Baseline: $(s_t^p) = (s_t^f)$, giving ℓ .
- Dove (Lubik and Schorfheide 2004): unreachable in baseline, active in counterfactuals.

Regime-specific rule

	ϕ_{π}	ϕ_x	ρ	$\widehat{P}_{kk}$
1: Eagle	3.18	0.47	0.97	0.79
2: S. Hawk	2.00	0.69	0.94	0.86
3: Hawk	1.54	0.50	0.89	0.84
4: W. Hawk	1.09	0.66	0.83	0.87
5: ZLB	0.13	0.02	0.86	0.84
6: Dove	0.77	0.68	0.60	0.98

Structural

$$\beta = 0.99 \quad \tau = 0.50 \quad \kappa = 0.086$$

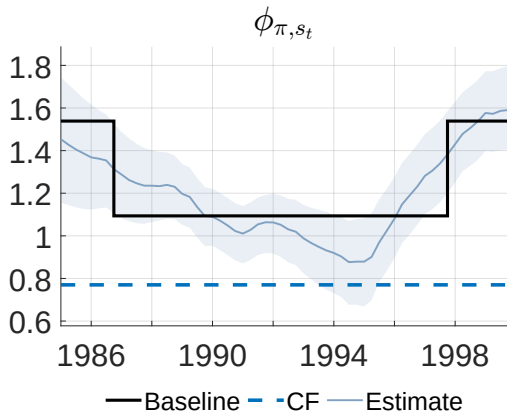
$$\rho_D = \rho_S = 0.85$$

ϕ_x annualized; $\rho = \rho_1 + \rho_2$ is total inertia.

Implication 2 – Source of the Great Moderation

Consensus: the Great Moderation followed a shift to a hawkish policy rule.

Clarida, Gali, and Gertler (2000), Coibion and Gorodnichenko (2011), and Bianchi (2013)

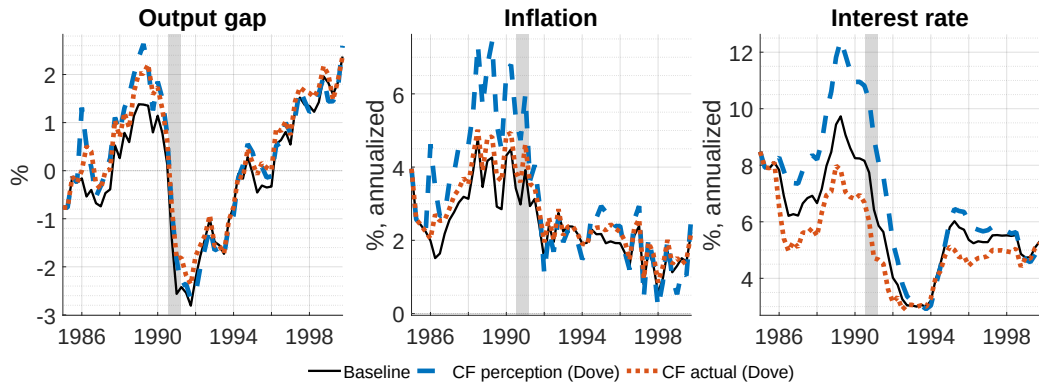


Baseline: Hawk **perception**, Hawk actual.

1. **Dove perception**, Hawk actual
 $\Rightarrow$ *Perception channel*
2. Hawk **perception**, **Dove** actual
 $\Rightarrow$ *Direct policy channel*

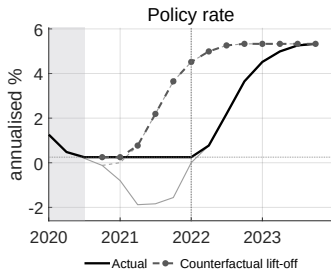
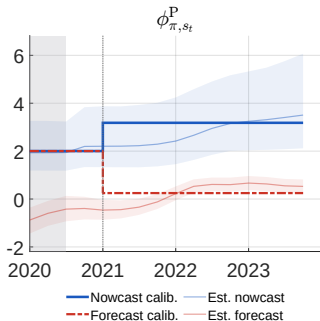
Q. What was the role of perception?

Implication 2 – Source of the Great Moderation



- Dovish beliefs amplify the level and volatility more than the Dovish rule.

Implication 3 – Misreading Perception (2021–2023)



Counterfactual: lift-off pulled at 2021Q1.

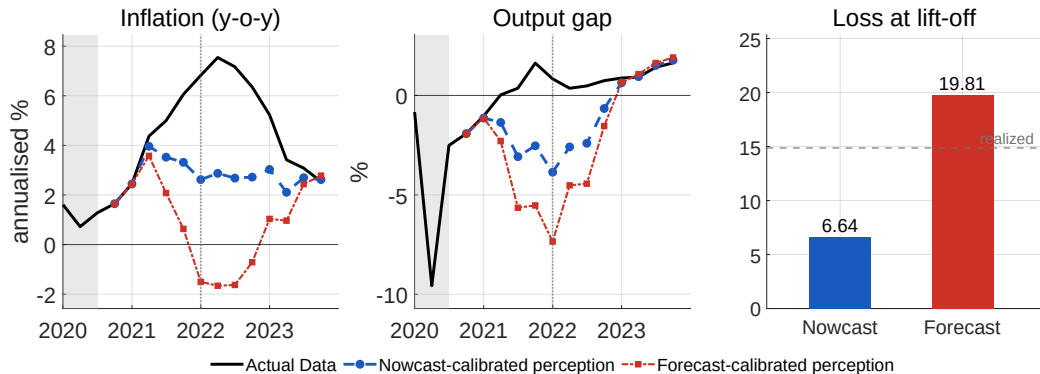
Two central banks read the *same* path:

1. **Nowcast**-calibrated ϕ_{π, s_t}^p
2. **Forecast**-calibrated ϕ_{π, s_t}^p

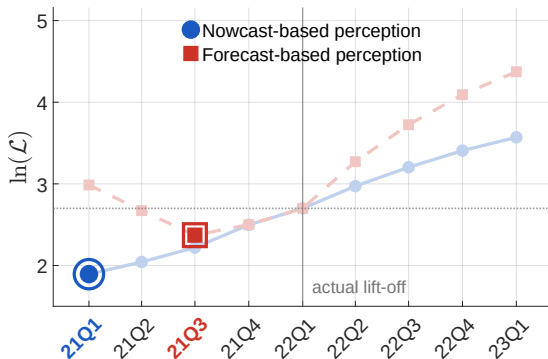
Each scores loss with:

$$\mathcal{L} = \mathbb{E} \left[\left(\pi_t^{\text{yoy}} - 2 \right)^2 + x_t^2 \right]$$

Implication 3 – Same Path, Different Readings



Implication 3 – The Cost of Misreading



- The same policy path is scored differently under the two **perceptions**.
- The **forecast**-based Fed rates the **nowcast**-optimal lift-off worse than its own optimal lift-off.
- Misreading **perception** leads to miscalculating policy outcome.

Conclusion and Contributions

1. Interest rate **forecasts** understate the **perceived** rule. **Nowcasts** recover it.
2. Determinacy depends on the **perceived** Taylor principle, not the actual one.
3. The Great Moderation required a shift in **perception**, not just policy.
4. Misreading **perception** may result in incorrect policy analysis.

*Reorienting the focus from the actual rule to the **perceived** rule.*